6-12 months were indirect and highly significant: ferritin – 165.25±70.45 *vs* 193.87±63.48 *vs* 141.91±18.65 (p=0.006) and iron – 77.67±33.95 *vs* 81.98±33.96 *vs* 78.83±30.11 (p=0.028). The ROC curve showed that ferritin seemed to be a good predictor for PAD unfavorable evolution (AUC = 0.603, IC95% 0.488-0.718, cut-off value – 128 mcg/ml).
Conclusions: According to the results, the ferritin evaluation in dynamics has importance in PAD prognostic predictability. However, for greater sensitivity, longer-term studies are needed that include a wider range of PAD-specific biomarkers analysis.

EP751 / #701, TOPIC: *ASA04 - CLINICAL VASCULAR DISEASE / ASA04-14 OTHER*, POSTER VIEWING SESSION.
ADRENAL ABNORMALITIES AND STROKE

H. Chaudhry [1,2], T. Baig [1,2], H. Kamel [1]. [1] *Neurology, Weill Cornell Medicine, New york, United States of America;* [2] *Neurology, Weill Cornell Medicine-Qatar, Doha, Qatar*

Background and Aims : Adrenal lesions are increasingly recognized as a potential cause of hyperaldosteronism and hypercortisolism, with adverse metabolic effects potentially increasing the risk of cardiovascular events. We examined the association of adrenal abnormalities with ischemic stroke and intracerebral hemorrhage (ICH).

Methods: We performed a cross-sectional study using data from the Cornell Acute Stroke Academic Registry (CAESAR), which includes patients with ischemic stroke and ICH at our medical center since 2018. As controls, we included randomly selected patients hospitalized with traumatic brain injury (TBI). We included patients who had undergone an abdominal CT with and without contrast. Imaging reports were reviewed for any mentions of adrenal abnormalities by reviewers blinded to the study hypothesis. We calculated prevalence rates of adrenal abnormalities by group and used logistic regression models, adjusted for demographics and vascular risk factors, to evaluate the association of adrenal abnormalities with ischemic stroke and ICH, in reference to those control patients with TBI.
Results: Among 75 IS patients, 75 ICH patients, and 75 TBI controls, adrenal abnormalities were found in 12% of the TBI patients, 13% of the ICH patients, and 15% of the IS patients (P = 0.89). In multiple logistic regression models, adrenal abnormalities were not associated with ischemic stroke (OR, 1.1; 95% CI, 0.4-3.0) or ICH (OR, 1.1; 95% CI, 0.4-2.9).
Conclusions: We found that patients with ischemic stroke or ICH did not have a higher prevalence of adrenal abnormalities than control patients. Future studies may be warranted to further investigate the role of occult hypercortisolism and hyperaldosteronism in stroke risk.

EP752 / #1463, TOPIC: *ASA04 - CLINICAL VASCULAR DISEASE / ASA04-14 OTHER*, POSTER VIEWING SESSION.
MACHINE LEARNING IN PREDICTION OF IN-STENT RESTENOSIS

A.Y. Filatova [1], V.I. Vybornov [2], E.A. Pylaeva [1], A.V. Potekhina [3], T.I. Arefieva [1], S.I. Provatorov [3]. [1] *Laboratory Of Cell Immunology, FSBO National Medical Research Center of Cardiology of Russian Ministry of Health, Moscow, Russian Federation;* [2] *Space Research Institute, Space Research Institute, Moscow, Russian Federation;* [3] *Department Of Pulmonary Hypertension And Heart Diseases, FSBO National Medical Research Center of Cardiology of Russian Ministry of Health, Moscow, Russian Federation, Moscow, Russian Federation*

Background and Aims : In-stent restenosis (ISR) after coronary angioplasty and stent implantation is associated with recurrent angina symptoms, higher risk for acute coronary syndrome, and increased mortality. Current ISR risk prediction models (PRESTO-1, PRESTO-2, and EVENT) suffer from insufficient predictive power. We aimed to investigate patient outcome prediction with classification predictive modeling in Machine Learning.

Methods: In retrospective study 203 available cases of ISR and 1009 control cases were included. Twenty seven demographic, clinical, biochemical and angiographic characteristics from each patient were obtained. Such machine-learning classifiers as logistic regression, random forest and XGBoost were used. The initial dataset was randomized into training (n=1057) and test (n=155) datasets. The prediction was performed by classifying samples into low or high digital risk score (DRS) groups. The prediction of outcome in the test dataset performed by machine-learning classifiers was further compared with the human expert classification.
Results: The incidence of ISR in the test dataset was 11%. Among the used machine-learning algorithms, the XGBoost classifier demonstrated the best predictive capacity in the unbalanced dataset. The DRS classification resulted in an odds ratio of 14.57 (95% CI 2.69–84.30, p=0.01) for prediction of ISR. The accuracy (C-index) of the DRS grouping was 0.89 (95% CI 0.84–0.94), as compared to 0.75 (95% CI 0.68–0.82) for human expert prediction in the same cohort.
Conclusions: Development of machine-learning algorithms will allow to reveal patients at higher risk of ISR for the further closer monitoring ISR or consideration of alternate therapies. A prospective validation is required before this biomarker can be implemented in clinical workflows.

EP753 / #582, TOPIC: *ASA04 - CLINICAL VASCULAR DISEASE / ASA04-14 OTHER*, POSTER VIEWING SESSION.
RELATIONSHIP BETWEEN BIOMARKERS AND GLOBAL MYOCARDIAL STRAIN PARAMETERS IN PATIENTS WITH CARDIOVASCULAR PATHOLOGY AND METABOLIC DISORDERS AFTER COVID-19-ASSOCIATED PNEUMONIA

T. Petelina [1], N. Musikhina [2], E. Yaroslavskaya [3], E. Gorbatenko [3], N. Galeeva [4], K. Avdeeva [1], A. Shcherbinina [1], L. Gapon [1]. [1] *Arterial Hypertension Department, Tyumen Cardiology Research Center, Tomsk National Research Medical Center, Russian Academy of Sciences, Tyumen, Russian Federation;* [2] *Urgent Cardiology Department, Tyumen Cardiology Research Center, Tomsk National Research Medical Center, Russian Academy of Sciences, Tyumen, Russian Federation;* [3] *Department Of Ultrasound Diagnostics, Tyumen Cardiology Research Center, Tomsk National Research Medical Center, Russian Academy of Sciences, Tyumen, Russian Federation;* [4] *Physiotherapeutic Treatment Department, Tyumen Cardiology Research Center, Tomsk National Research Medical Center, Russian Academy of Sciences, Tyumen, Russian Federation*

Background and Aims : To study dynamics of biomarkers of inflammatory response and to estimate their association with parameters of ventricular systolic function in arterial hypertension (AH) and coronary artery disease (CAD) patients with and without metabolic disorders (MD) such as obesity and impaired carbohydrate metabolism of those who underwent COVID-19-associated pneumonia.

Methods: The study included 264 patients (56.97±8.48 years). Gr.1 (n=116) involved AH and CAD patients without MD, gr.2 (n=148) AH and CAD patients with MD. Age, volume of lung damage were comparable. Baseline parameters of complete blood count, biochemistry were assessed on the day of hospitalization. In-depth analysis of biomarkers and echocardiography with speckle tracking were performed 3 months after.
Results: Initially in both groups, statistically significant exaggeration of the normal range of maximum CRP; NLR; PLR; ESR were registered. Dynamics of CRP, NLR, PLR, ESR reached normative values in both groups after 3 months, however, hs-CRP was 4.75±2.09 and 7.06±3.13 mg/l with exaggeration of value in gr.2 (p<0.001). Global myocardial strain parameters were significantly more apparent in gr.2. In both groups, parameters of the strain were positively correlated with parameters of RBC, HGB, HCT, creatinine, APTT. Additionally, relationship with LYM was revealed in gr.1, positive relationship with CRP, TG, GFR, and negative with HDL-C was found in gr.1.
Conclusions: In group of patient with AH, CAD and MD after COVID-19, more significant alteration was detected in global myocardial strain associated with changed parameters of hematological and inflammatory biomarkers, correction of which can lead to optimization of contractile ventricular function.